American Computer Science League

2022 Finals ● Short Problems ● Senior Division

1. Boolean Algebra

Simplify the following expression:

$$A(\overline{A} + B)\,\overline{C} + \overline{A\,C} + \overline{B + C}$$

A. $\overline{A} + C$
B. $\overline{A} + \overline{C}$
C. $A + \overline{C}$
D. $A + C$
E. None of the above

2. Boolean Algebra

How many ordered triples make the following expression FALSE?

$$\overline{A}\,(\overline{A + B})\,(\overline{A\,B})\,(\overline{A}\,B)\,\overline{C}$$

A. 0
B. 2
C. 4
D. 8
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

((NOT (LCIRC-1 10110) OR (RSHIFT-2 01100))
AND (LSHIFT-1 (RCIRC-2 01101)))

A. 10010
B. 10110
C. 10011
D. 00011
E. None of the above

4. Bit-String Flicking

How many bit string values of X (a bitstring of 5 bits) make the following equation TRUE?

((LCIRC-3 (10100 OR X)) AND (11010 OR (RSHIFT-1 X))) = 11001

A. 0
B. 2
C. 4
D. 6
E. None of the above

5. Recursive Functions

Find $f(f(f(f(20))))$ given:

$$f(x) = \begin{cases} f(x-2) + 1 & \text{if } x \text{ is odd and prime} \\ f\left(\left[\frac{x}{2}\right]\right) + 3 & \text{if } x \text{ is even} \\ x + 1 & \text{otherwise} \end{cases}$$

Remember that [x] = greatest integer less than or equal to x.

A. 4
B. 5
C. 7
D. 10
E. None of the above

6. Recursive Functions

Define Pisano's function as

$$P(n) = \begin{cases} 0 & if\ n = 1 \\ 1 & if\ n = 2 \\ (P(n-1) + P(n-2))\ mod\ m & if\ n \geq 3 \end{cases}$$

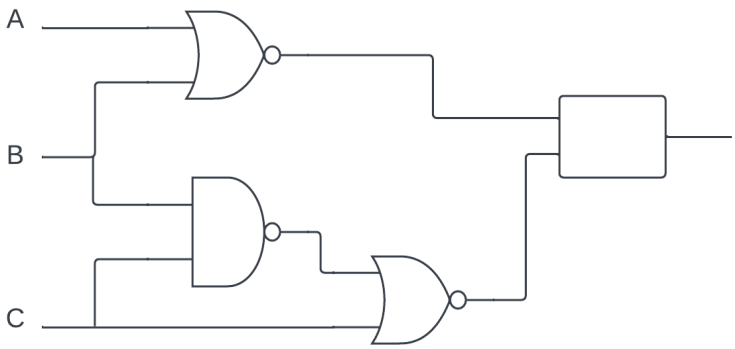
The number of terms that repeat is called the Pisano period. For *mod* 2 the sequence becomes 0, 1, 1, 0, 1, 1, … so the Pisano period for *mod* 2 is 3.

What is the Pisano period for *mod* 4 (i.e. m = 4)?

A. 4
B. 6
C. 8
D. 10
E. None of the above

7. Digital Electronics

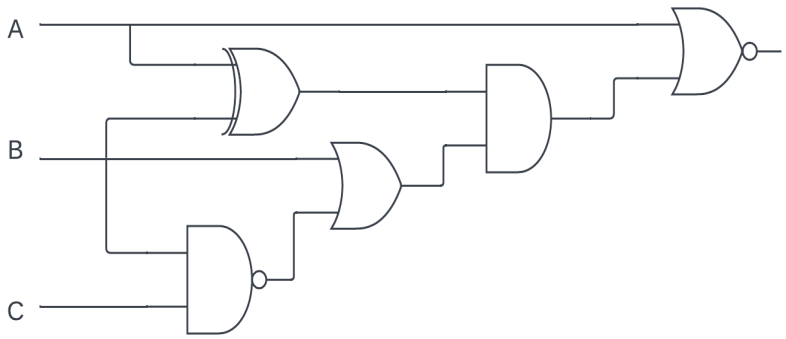
What gate needs to replace the rectangle to make all outputs FALSE?



A. AND
B. OR
C. NAND
D. NOR
E. None of the above

8. Digital Electronics

Simplify the Boolean expression represented by the digital circuit below:



A. $A + B$
B. $\overline{A} + B$
C. $A + \overline{B}$
D. $\overline{A + B}$
E. None of the above

9. Prefix-Infix-Postfix

Evaluate the following prefix expression if # = max(a,b,c) and $ = min(a,b,c). Note: numbers are single digits.

+ + / * # 1 2 3 4 6 * $ 8 4 2 9 # 3 $ 4 5 6 2

A. 28
B. 26
C. 24
D. 22
E. None of the above

10. Prefix-Infix-Postfix

Given the expression written in infix, convert it to postfix:

$$\frac{1}{x + \frac{1}{x + \frac{1}{x+1}}}$$

A. 1 x 1 x 1 x 1 + / + / + /
B. 1 x 1 x 1 x 1 + / + / / +
C. 1 1 1 1 x x x + / + / + /
D. 1 x 1 x 1 x 1 + + + / / /
E. None of the above

11. Computer Number Systems

When is the next year (in base10) after 2022_{10} that the year expressed in hexadecimal will have 3 consecutive hexadecimal digits in increasing or decreasing order?

A. 2102
B. 2066
C. 2202
D. 2166
E. None of the above

12. Computer Number Systems

In the octal representation for the decimal numbers from 50 to 100 inclusive, how many times does the digit 2 occur?

A. 13
B. 14
C. 15
D. 16
E. None of the above

13. Data Structures

Build a binary search tree for: **OMICRONVARIANT**

What is the internal path length of the tree?

A. 35
B. 36
C. 37
D. 38
E. None of the above

14. Data Structures

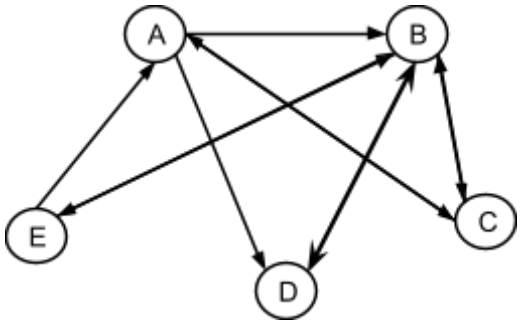
Perform the following commands on an initially empty stack and on an initially empty queue, read left to right first. Determine what the next item popped will be on each. Which of these comes last in alphabetical order?

PUSH("A"),	X=POP(),	PUSH("D"),	PUSH("V"),
PUSH("E"),	X=POP(),	X=POP(),	PUSH("N"),
PUSH("T"),	PUSH("U"),	PUSH("R"),	X=POP(),
X=POP(),	PUSH("O"),	PUSH("U"),	PUSH("S"),
X=POP(),	X=POP(),	X=POP(),	X=POP(),
X=POP()			

A. D
B. S
C. N
D. U
E. No difference

15. Graph Theory

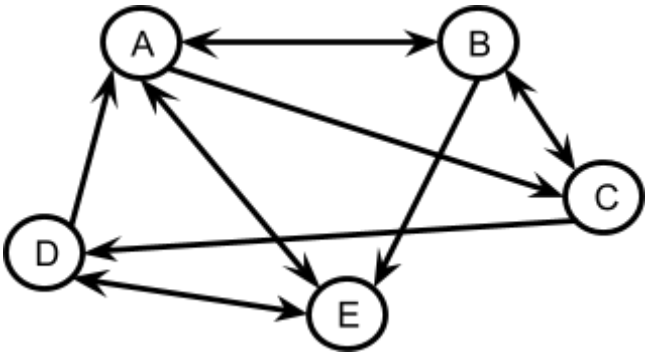
Given the directed graph below, what is the total number of paths of length 2?



A. 24
B. 25
C. 26
D. 27
E. None of the above

16. Graph Theory

How many cycles of length 4 are there in the directed graph below?



A. 2
B. 3
C. 4
D. 5
E. None of the above

17. What Does This Program Do?

Given the array **fib** that contains the Fibonacci numbers 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584, 4181 in locations 0 to 19, determine the output for the following program:

```
cnt = 0
for x = 3 to 6
    f = fib(x)
    for y = x+1 to 19
        if fib(y) % f == 0 then
            cnt = cnt + 1
        end if
    next y
next x
output cnt
```

A. 12
B. 11
C. 10
D. 9
E. None of the above

18. LISP

Evaluate the final LISP expression.

```
(SETQ X  '((1 2 (3)) (2) 3 (1 (2 (3)))))
(SETQ Y  '(2 1 (3 (3))))
(SETQ Z (CONS Y X))
(SETQ W (REVERSE Z))
(CADADAR W)
```

A. (2 (3))
B. ((3))
C. 3
D. (3)
E. None of the above

19. FSA and Regular Expressions

The manager of the ACSL Garden Center needs to place an order to have enough inventory for the Grand Opening. She may order only items that satisfy the following regular expression:

[^iop] [aeiou] [a-g]* [m-t]* (e|s)*

How many items can she order?

mulch	seeds	rake	weeder
daylily	ivy	pots	berries
mower	grass	hoses	straw

A. 2
B. 3
C. 4
D. 5
E. None of the above

20. Assembly Language

What is printed when the following program is run if the data is the pair of decreasing Fibonacci numbers 8 and 5?

```
T       DC      0
        READ    N3
        READ    N2
        LOAD    N3
        ADD     N2
        MULT    N3
        STORE   P
LOOP    LOAD    N3
        SUB     N2
        STORE   N1
        LOAD    N3
        MULT    N3
        ADD     T
        STORE   T
        LOAD    N2
        STORE   N3
        LOAD    N1
        STORE   N2
        BG      LOOP
        LOAD    P
        SUB     T
        STORE   ANS
        PRINT   ANS
        END
```

A. 0
B. 1
C. 2
D. 3
E. None of the above